

# FUNCTIONAL REQUIREMENTS DOCUMENT

## Module: Knome

*Knowledge Sharing | Communities | Technical Blog | Social Networking*

Inhouse Application Development — Enterprise Excellence Portal (EEP)

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| Project Name    | Knome                                                                                |
| Module          | Knome (Knowledge Sharing, Group & Technical Post)<br>(Technical Social media, Blogs) |
| Document Type   | Functional Requirement Document (FRD)                                                |
| Version         | 1.0                                                                                  |
| Status          | <b>DRAFT — PENDING REVIEW</b>                                                        |
| Prepared By     | Nilesh Wadbude                                                                       |
| Reviewed By     | Deven Goratela (Subject Matter Expert / Senior)                                      |
| Date            | 20 June 2026                                                                         |
| Confidentiality | <b>CONFIDENTIAL — Internal Use Only</b>                                              |

मध्यप्रदेश सरकार का पोर्टल

**MPonline Limited**  
Transforming Governance. Empowering Citizens.  
Joint venture between Govt of Madhya Pradesh and TCS



# 1. Document Control

## 1.1 Revision History

| Version | Date        | Author         | Description                                 | Status |
|---------|-------------|----------------|---------------------------------------------|--------|
| 1.0     | 22-Jun-2026 | Nilesh Wadbude | Initial Draft — FRD creation from Pre-Notes | Draft  |

## 1.2 Meeting Summary

| Attribute    | Details                                                                                     |
|--------------|---------------------------------------------------------------------------------------------|
| Meeting Date | 10 Jun 2026                                                                                 |
| Duration     | 30 minutes 20 seconds                                                                       |
| Participants | Deven Goratela (SME/Senior), Nilesh Wadbude (Team Member), Deepak Karma (Team Member)       |
| Application  | Inhouse Application Development — Module: Knome (Knowledge Sharing, Group & Technical Post) |
| Prepared By  | Nilesh Wadbude                                                                              |

## 1.3 Approvals

| Role               | Name                                   | Signature | Date         |
|--------------------|----------------------------------------|-----------|--------------|
| Document Author    | Nilesh Wadbude                         |           | 20 June 2026 |
| Technical Reviewer | Deven Goratela (Subject Matter Expert) |           | 24 June 2026 |
| Module Owner       |                                        |           |              |
| Project Manager    | Gagan Soni (CTO)                       |           | 24 June 2026 |

## 2. Introduction

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### 2.1 Purpose

This Functional Requirements Document (FRD) defines the comprehensive functional specifications, business rules, role-based access controls, use cases, and non-functional requirements for the Knome module — an enterprise social networking and knowledge-sharing platform developed as part of MPOnline Limited's Enterprise Excellence Portal (EEP) initiative.

This document serves as the contractual baseline between the Business Stakeholders, Solution Architects, and Development Team, governing design, development, testing, and deployment decisions.

### 2.2 Scope

The Knome module encompasses the following functional domains:

- Dashboard and personalized content feed
- User profile management and professional networking
- Content creation: posts, articles/blogs, videos, and podcasts
- Community management (public, private, and default communities)
- Internal job posting board
- Karma Points engagement engine
- Search and discovery across all content types
- Analytics dashboards for admins and HR
- Content moderation and security controls
- Integration with HRMS, Active Directory/SSO, and Microsoft ecosystem

#### Out of Scope (Phase 1):

- AI content recommendation engine (Phase 2)
- Live streaming capabilities (Phase 2)
- External social media integration
- Mentorship network and Learning Management integration

### 2.3 Background

MPOnline Limited, as a state government IT service provider for Madhya Pradesh, requires an internal enterprise collaboration platform to foster knowledge sharing across departments, improve cross-functional communication, and create an inclusive environment for organizational learning and professional growth.

The Knome platform is inspired by leading enterprise social platforms combining features of LinkedIn (professional networking), Medium/Blogger (blogs and articles), YouTube (video channel), Podcast platforms, and internal community forums — tailored specifically for the organizational culture and compliance requirements of a government IT organization.

## 2.4 Intended Audience

| Audience                                | Purpose                                              |
|-----------------------------------------|------------------------------------------------------|
| Business Analysts / Solution Architects | Baseline requirements definition and solution design |
| Development Team                        | Technical implementation reference                   |
| QA / Testing Team                       | Test case preparation and acceptance criteria        |
| HR / Business Owners                    | Feature validation and sign-off                      |
| Project Management                      | Scope control and delivery tracking                  |

## 3. Business Objectives

The Knome module is designed to achieve the following measurable business objectives:

| Obj. ID | Objective                                                       | Success Metric                                      |
|---------|-----------------------------------------------------------------|-----------------------------------------------------|
| OBJ-01  | Promote organizational knowledge sharing across all departments | 50%+ employees active monthly within 6 months       |
| OBJ-02  | Enable employees to create and publish diverse content types    | 100+ posts / articles published per month           |
| OBJ-03  | Foster collaboration through community-based engagement         | 10+ active communities with 5+ members each         |
| OBJ-04  | Encourage engagement through Karma Points gamification          | 80%+ of active users earning points monthly         |
| OBJ-05  | Support internal talent mobility via job board                  | All open internal vacancies posted on platform      |
| OBJ-06  | Improve cross-departmental communication and engagement         | 25%+ reduction in email-based internal broadcasting |

## 4. User Roles and Responsibilities

### 4.1 Role Summary

| Role Code     | Role Name            | Description                                                                              |
|---------------|----------------------|------------------------------------------------------------------------------------------|
| <b>EMP</b>    | Employee             | All organization employees who consume and create content on the platform                |
| <b>CADM</b>   | Community Admin      | Designated manager of a specific community; moderates and governs community content      |
| <b>HRADM</b>  | HR Administrator     | HR team managing platform-level configurations, announcements, and engagement monitoring |
| <b>SYSADM</b> | System Administrator | Technical admin responsible for platform configuration, access management, and analytics |

### 4.2 Role Capability Matrix

| Feature / Action                      | Employee | Community Admin | HR Admin | Sys Admin |
|---------------------------------------|----------|-----------------|----------|-----------|
| Create / Edit personal profile        | ✓        | ✓               | ✓        | ✓         |
| Create text / image / video post      | ✓        | ✓               | ✓        | ✗         |
| Publish articles / blogs              | ✓        | ✓               | ✓        | ✗         |
| Upload video to Video Channel         | ✓        | ✓               | ✓        | ✗         |
| Upload / record podcast               | ✓        | ✓               | ✓        | ✗         |
| Create community                      | ✓        | ✗               | ✓        | ✓         |
| Approve community join requests       | ✗        | ✓               | ✓        | ✓         |
| Moderate community content            | ✗        | ✓               | ✓        | ✓         |
| Manage platform banners/announcements | ✗        | ✗               | ✓        | ✓         |
| Post internal job vacancies           | ✗        | ✗               | ✓        | ✓         |
| View analytics dashboard              | ✗        | ✓               | ✓        | ✓         |
| Manage user access and roles          | ✗        | ✗               | ✗        | ✓         |
| Configure system settings             | ✗        | ✗               | ✗        | ✓         |
| Earn / view Karma Points              | ✓        | ✓               | ✓        | ✗         |

|                                  |   |   |   |   |
|----------------------------------|---|---|---|---|
| Follow employees / build network | ✓ | ✓ | ✓ | X |
| Report / flag content            | ✓ | ✓ | ✓ | ✓ |
| Delete / suspend users           | X | X | X | ✓ |

## 5. Functional Requirements

### 5.1 Dashboard & Personalized Feed

The Dashboard is the primary landing page presenting personalized content from the user's network, communities, and platform-wide trends.

| Req ID   | Requirement Description                                                                                                                                | Priority | Type        | MoSCoW      | Notes                 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|-------------|-----------------------|
| FR-DB-01 | System shall display a personalized timeline/feed based on user's community memberships and followed employees                                         | High     | Functional  | Must Have   | —                     |
| FR-DB-02 | System shall display trending/hot posts calculated by engagement score algorithm                                                                       | High     | Functional  | Must Have   | Algo defined in §5.11 |
| FR-DB-03 | System shall display community suggestions based on user's department and interests                                                                    | Medium   | Functional  | Should Have | —                     |
| FR-DB-04 | System shall display people suggestions based on common department, skills, and mutual connections                                                     | Medium   | Functional  | Should Have | —                     |
| FR-DB-05 | System shall display organization-wide announcements from HR Administrator                                                                             | High     | Functional  | Must Have   | —                     |
| FR-DB-06 | Dashboard shall contain widgets: Communities Joined, Suggested Communities, Hot Posts, Latest Articles, Videos, Podcasts, Internal Jobs, Karma Ranking | High     | Functional  | Must Have   | —                     |
| FR-DB-07 | User shall be able to filter the feed by: All Posts, Communities, Articles, Videos, Podcasts, Jobs                                                     | Medium   | Functional  | Must Have   | —                     |
| FR-DB-08 | Dashboard shall load within 3 seconds for authenticated users on a standard connection                                                                 | High     | Performance | Must Have   | See NFR-PERF-01       |

## 5.2 User Profile Management

Each employee has a rich profile page that serves as their professional identity and content portfolio within the platform.

| Req ID   | Requirement Description                                                                                                                             | Priority | Type        | MoSCoW      | Notes     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|-------------|-----------|
| FR-UP-01 | System shall auto-populate user profile with data from HRMS: Name, Designation, Department, Location                                                | High     | Integration | Must Have   | HRMS sync |
| FR-UP-02 | User shall be able to update Bio, Skills, Interests, and upload a profile photo                                                                     | High     | Functional  | Must Have   | —         |
| FR-UP-03 | Profile page shall display sections: About Me, My Posts, My Articles, My Videos, My Podcasts, Communities Joined, Followers/Following, Karma Points | High     | Functional  | Must Have   | —         |
| FR-UP-04 | User shall be able to configure visibility settings for Bio, Network, Photos, and Interests (Public / Connections Only / Private)                   | Medium   | Functional  | Should Have | —         |
| FR-UP-05 | System shall display follower count, following count, and mutual connections on profile page                                                        | Medium   | Functional  | Must Have   | —         |
| FR-UP-06 | System shall support profile photo upload with auto-resize to platform standard dimensions                                                          | Low      | Functional  | Should Have | —         |

## 5.3 Post Creation

Employees can create quick-share posts supporting text, images, documents, and media attachments.

| Req ID   | Requirement Description                                                                                         | Priority | Type       | MoSCoW    | Notes                 |
|----------|-----------------------------------------------------------------------------------------------------------------|----------|------------|-----------|-----------------------|
| FR-PC-01 | User shall be able to create text posts up to 400 characters with optional media attachments                    | High     | Functional | Must Have | Hard limit: 400 chars |
| FR-PC-02 | Post shall support attachment types: Image (JPG/PNG/GIF), Document (PDF/DOCX), Video (MP4/MOV), Audio (MP3/WAV) | High     | Functional | Must Have | —                     |
| FR-PC-03 | User shall be able to set audience for post: Everyone, Specific Communities, or Specific Users                  | High     | Functional | Must Have | —                     |
| FR-PC-04 | System shall support post actions: Publish Immediately, Schedule for Later, Save as Draft, Cancel               | High     | Functional | Must Have | —                     |

|                 |                                                                                                |               |            |             |           |
|-----------------|------------------------------------------------------------------------------------------------|---------------|------------|-------------|-----------|
| <b>FR-PC-05</b> | Post character limit indicator shall update in real-time as user types                         | <b>Medium</b> | Functional | Should Have | —         |
| <b>FR-PC-06</b> | Users shall be able to @mention other employees in posts; mentioned user receives notification | <b>Medium</b> | Functional | Should Have | —         |
| <b>FR-PC-07</b> | System shall validate URLs in posts for malicious or restricted content before publishing      | <b>High</b>   | Security   | Must Have   | See §5.15 |

## 5.4 Articles and Blogs

The article editor enables rich long-form content creation comparable to enterprise blogging platforms.

| Req ID          | Requirement Description                                                                                                               | Priority      | Type        | MoSCoW      | Notes |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|-------------|-------|
| <b>FR-AB-01</b> | User shall be able to create articles with: Title, Description, Rich Text Content, Category, and Tags                                 | <b>High</b>   | Functional  | Must Have   | —     |
| <b>FR-AB-02</b> | Rich text editor shall support: Bold, Italics, Underline, Tables, Inline Images, Hyperlinks, Bulleted Lists, Numbered Lists, Headings | <b>High</b>   | Functional  | Must Have   | —     |
| <b>FR-AB-03</b> | Articles shall support multi-media attachments: Documents, Images, Videos, Podcasts                                                   | <b>Medium</b> | Functional  | Should Have | —     |
| <b>FR-AB-04</b> | System shall support article actions: Publish, Schedule, Save Draft, Cancel                                                           | <b>High</b>   | Functional  | Must Have   | —     |
| <b>FR-AB-05</b> | Published articles shall be indexed and searchable within 5 minutes of publication                                                    | <b>Medium</b> | Performance | Should Have | —     |
| <b>FR-AB-06</b> | System shall track article view count, unique reads, average read time, and reactions                                                 | <b>Medium</b> | Functional  | Should Have | —     |
| <b>FR-AB-07</b> | Author shall be able to edit or delete their own articles post-publication with version history maintained                            | <b>Medium</b> | Functional  | Should Have | —     |

## 5.5 Video Channel

The Video Channel provides an internal YouTube-like capability for organizational knowledge and training videos.

| Req ID   | Requirement Description                                                                               | Priority | Type       | MoSCoW      | Notes                    |
|----------|-------------------------------------------------------------------------------------------------------|----------|------------|-------------|--------------------------|
| FR-VC-01 | User shall be able to upload videos with Title, Description, Tags, Category, and Custom Thumbnail     | High     | Functional | Must Have   | —                        |
| FR-VC-02 | System shall support video sources: Direct upload, OneDrive link, Microsoft Stream link, Embedded URL | High     | Functional | Must Have   | MS ecosystem integration |
| FR-VC-03 | Video player shall support standard controls: play/pause, seek, volume, full screen, playback speed   | High     | Functional | Must Have   | —                        |
| FR-VC-04 | System shall display view count, likes, and comments per video                                        | Medium   | Functional | Must Have   | —                        |
| FR-VC-05 | Videos shall support categorization and tag-based discovery                                           | Medium   | Functional | Should Have | —                        |
| FR-VC-06 | Maximum direct upload file size: 500MB; supported formats: MP4, MOV, AVI, MKV                         | High     | Functional | Must Have   | —                        |

## 5.6 Podcast Channel

| Req ID   | Requirement Description                                                                        | Priority | Type       | MoSCoW      | Notes |
|----------|------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-PD-01 | User shall be able to upload audio files or record audio directly in-browser for podcasts      | High     | Functional | Must Have   | —     |
| FR-PD-02 | Podcast entries shall include: Title, Description, Duration, Category, Cover Image             | High     | Functional | Must Have   | —     |
| FR-PD-03 | System shall support podcast series grouping where multiple episodes are linked under a series | Medium   | Functional | Should Have | —     |
| FR-PD-04 | Audio player shall support: play/pause, seek, speed adjustment (0.5x–2x), volume               | High     | Functional | Must Have   | —     |
| FR-PD-05 | Supported audio formats: MP3, WAV, AAC, OGG. Maximum upload size: 100MB                        | High     | Functional | Must Have   | —     |

## 5.7 Community Management

Communities serve as the primary collaboration hubs. The platform supports multiple community types with distinct governance models.

| Req ID   | Requirement Description                                                                                                           | Priority | Type       | MoSCoW      | Notes        |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|----------|------------|-------------|--------------|
| FR-CM-01 | System shall support three community types: Public (open join), Private (request to join), Default/Organization (auto-subscribed) | High     | Functional | Must Have   | —            |
| FR-CM-02 | Community creation shall require: Name, Description, Banner Image, Thumbnail, Category, Rules, FAQ, Community Type                | High     | Functional | Must Have   | —            |
| FR-CM-03 | For private communities, Community Admin shall approve or reject join requests                                                    | High     | Functional | Must Have   | —            |
| FR-CM-04 | HR Admin shall be able to create default communities auto-assigned to all employees or by department                              | High     | Functional | Must Have   | —            |
| FR-CM-05 | Default communities shall include: HR, Finance, Technology, CTO, Marketing                                                        | Medium   | Functional | Must Have   | Configurable |
| FR-CM-06 | Community shall support two member types: Subscriber (view-only) and Member (can post and comment)                                | Medium   | Functional | Must Have   | —            |
| FR-CM-07 | Community Admin shall be able to moderate content: pin posts, remove posts, suspend members                                       | High     | Functional | Must Have   | —            |
| FR-CM-08 | Community page shall display: member count, recent posts, pinned content, community rules, FAQ, Admin contact                     | Medium   | Functional | Should Have | —            |
| FR-CM-09 | Users shall be able to perform community actions: Create, Join, Subscribe, Leave                                                  | High     | Functional | Must Have   | —            |

## 5.8 Internal Job Posting Board

| Req ID   | Requirement Description                                                                                                                                    | Priority | Type        | MoSCoW      | Notes |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|-------------|-------|
| FR-JB-01 | HR Admin shall be able to create internal job postings with: Job Title, Department, Description, Skills Required, Location, Closing Date, Application Link | High     | Functional  | Must Have   | —     |
| FR-JB-02 | Jobs shall be filterable by Department, Location, Skills, and Expiry Date on the dashboard                                                                 | Medium   | Functional  | Should Have | —     |
| FR-JB-03 | System shall auto-expire and archive job postings after the specified closing date                                                                         | High     | Functional  | Must Have   | —     |
| FR-JB-04 | Employees shall receive bell notification when new jobs matching their department/skills are posted                                                        | Medium   | Functional  | Could Have  | —     |
| FR-JB-05 | Job posting application link shall redirect to HRMS or external ATS for application processing                                                             | Medium   | Integration | Should Have | —     |

## 5.9 People Network

| Req ID   | Requirement Description                                                                                            | Priority | Type       | MoSCoW      | Notes |
|----------|--------------------------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-PN-01 | User shall be able to follow any employee; following is asymmetric (no mutual acceptance required)                 | High     | Functional | Must Have   | —     |
| FR-PN-02 | System shall suggest 'People You May Know' based on department, skills, mutual connections, and common communities | Medium   | Functional | Should Have | —     |
| FR-PN-03 | Profile shall display follower count, following count, mutual connection count, and common communities             | Medium   | Functional | Must Have   | —     |
| FR-PN-04 | User shall be able to unfollow at any time; unfollowed user is not notified                                        | Medium   | Functional | Must Have   | —     |

## 5.10 Content Interactions

| Req ID   | Requirement Description                                                                           | Priority | Type       | MoSCoW      | Notes |
|----------|---------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-CI-01 | Users shall be able to react to any post, article, or video with: Like, Celebrate, Support, Heart | High     | Functional | Must Have   | —     |
| FR-CI-02 | Users shall be able to add comments with text and optional image attachment on all content types  | High     | Functional | Must Have   | —     |
| FR-CI-03 | Users shall be able to share content to: their timeline, specific communities, or specific users  | Medium   | Functional | Must Have   | —     |
| FR-CI-04 | Users shall be able to save/bookmark content; Saved items accessible from profile                 | Medium   | Functional | Should Have | —     |
| FR-CI-05 | Comment threads shall support nested replies up to 2 levels deep                                  | Medium   | Functional | Should Have | —     |

## 5.11 Hot Posts Ranking Engine

The Hot Posts feature surfaces the most engaging content using a time-weighted engagement scoring algorithm.

| Req ID   | Requirement Description                                                                                                                             | Priority | Type        | MoSCoW      | Notes                |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|-------------|----------------------|
| FR-HP-01 | System shall calculate engagement score per post using formula: $(Views \times 1) + (Reactions \times 3) + (Comments \times 5) + (Shares \times 4)$ | High     | Functional  | Must Have   | Weights configurable |
| FR-HP-02 | System shall maintain Hot Posts rankings for three time windows: Daily, Weekly, Monthly                                                             | High     | Functional  | Must Have   | —                    |
| FR-HP-03 | Hot Posts widget on Dashboard shall display top 5 posts by current daily engagement score                                                           | Medium   | Functional  | Must Have   | —                    |
| FR-HP-04 | Hot Posts rankings shall recalculate every 15 minutes                                                                                               | Medium   | Performance | Should Have | —                    |

## 5.12 Notifications

| Req ID   | Requirement Description                                                                                                                                                                  | Priority | Type       | MoSCoW      | Notes |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-NT-01 | System shall send real-time bell notifications for: New Comment on my content, New Reaction on my content, New Follower, Community Invitation, @Mention, New Community Post, Job Posting | High     | Functional | Must Have   | —     |
| FR-NT-02 | System shall send email digest notifications: Daily Digest (summary of activity), Weekly Digest (top content)                                                                            | Medium   | Functional | Should Have | —     |
| FR-NT-03 | User shall be able to configure notification preferences per event type (Enable/Disable Bell, Email)                                                                                     | High     | Functional | Must Have   | —     |
| FR-NT-04 | Notification centre shall display unread count badge and mark-all-as-read option                                                                                                         | Medium   | Functional | Must Have   | —     |

## 5.13 Karma Points Engagement System

The Karma Points system gamifies knowledge contribution and drives ongoing engagement through a structured reward and recognition model.

| Activity                       | Points Awarded | Frequency Cap                      |
|--------------------------------|----------------|------------------------------------|
| Create and publish a Post      | 2 pts          | Max 10 pts/day from posts          |
| Publish an Article             | 10 pts         | Max 30 pts/day from articles       |
| Upload a Video                 | 8 pts          | Max 24 pts/day from videos         |
| Upload a Podcast               | 8 pts          | Max 24 pts/day from podcasts       |
| Receive a Like on content      | 1 pt           | No cap — accumulates from audience |
| Receive a Comment on content   | 2 pts          | No cap                             |
| Receive a Share on content     | 3 pts          | No cap                             |
| Active Community Participation | 5 pts/day      | Once per community per day         |

| Req ID   | Requirement Description                                                                                         | Priority | Type       | MoSCoW      | Notes |
|----------|-----------------------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-KP-01 | System shall automatically award Karma Points based on the defined activity-point matrix                        | High     | Functional | Must Have   | —     |
| FR-KP-02 | System shall maintain a real-time Leaderboard displaying Top 10 contributors by Karma Points                    | High     | Functional | Must Have   | —     |
| FR-KP-03 | System shall award badges at defined Karma thresholds: Bronze (100), Silver (500), Gold (1000), Platinum (5000) | Medium   | Functional | Should Have | —     |
| FR-KP-04 | User's Karma Points history and breakdown shall be viewable on their profile                                    | Medium   | Functional | Should Have | —     |

## 5.14 Search and Discovery

| Req ID   | Requirement Description                                                                                      | Priority | Type        | MoSCoW      | Notes           |
|----------|--------------------------------------------------------------------------------------------------------------|----------|-------------|-------------|-----------------|
| FR-SD-01 | System shall provide a unified search covering: People, Communities, Posts, Articles, Videos, Podcasts, Jobs | High     | Functional  | Must Have   | —               |
| FR-SD-02 | Search results shall be filterable by: Date Range, Category, Department, Popularity (Most Viewed/Liked)      | High     | Functional  | Must Have   | —               |
| FR-SD-03 | Search shall support partial match, tag-based, and author-based search                                       | Medium   | Functional  | Should Have | —               |
| FR-SD-04 | Search results shall return results within 2 seconds for queries up to 50 characters                         | High     | Performance | Must Have   | See NFR-PERF-02 |
| FR-SD-05 | System shall display search history (last 10 searches) for logged-in users                                   | Low      | Functional  | Could Have  | —               |

## 5.15 Content Security and Moderation

| Req ID   | Requirement Description                                                                                                                                                                                                                                                                      | Priority | Type       | MoSCoW      | Notes |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|-------------|-------|
| FR-SM-01 | System shall perform real-time URL validation against a restricted/malicious URL database before post publication                                                                                                                                                                            | High     | Security   | Must Have   | —     |
| FR-SM-02 | Any user shall be able to report content with a reason code; reported content queued for moderation review                                                                                                                                                                                   | High     | Functional | Must Have   | —     |
| FR-SM-03 | Moderators (Community Admin / Sys Admin) shall be able to: Block, Remove, or Escalate reported content                                                                                                                                                                                       | High     | Functional | Must Have   | —     |
| FR-SM-04 | System shall support user suspension: temporary (with auto-reinstate date) or permanent                                                                                                                                                                                                      | High     | Functional | Must Have   | —     |
| FR-SM-05 | System shall maintain a complete audit trail of moderation actions with timestamp and moderator ID                                                                                                                                                                                           | High     | Compliance | Must Have   | —     |
| FR-SM-06 | Define the rules to limit the characters for Post and Article. They cannot be of infinite length. Also we need to add the point that each content which is posted on this platform has to be moderated so that no one posts illegal or marketing or self-promoting content on this platform, | High     | Functional | Must Have   | -     |
| FR-SM-07 | System shall implement content filtering to block predefined restricted keywords                                                                                                                                                                                                             | Medium   | Security   | Should Have | —     |

## 6. Key Use Cases

### 6.1 UC-01: Employee Publishes a Blog Article

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-01                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Use Case Name</b>    | Employee Publishes a Blog Article                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Actor(s)</b>         | Employee (Primary), Notification Service (Secondary)                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pre-condition(s)</b> | 1. Employee is authenticated via SSO. 2. Employee has 'Member' role (not suspended). 3. Employee has opened the Articles editor.                                                                                                                                                                                                                                                                                                          |
| <b>Basic Flow</b>       | 1. Employee clicks 'Create Article'. 2. Employee enters Title, selects Category, adds Tags. 3. Employee writes content using Rich Text Editor. 4. Employee optionally attaches media. 5. Employee clicks 'Publish'. 6. System validates content (URL check, character limits). 7. System publishes article; awards 10 Karma Points to author. 8. System indexes article for search. 9. Followers of the author receive bell notification. |
| <b>Alternate Flow</b>   | Step 5a: Employee clicks 'Schedule' — system prompts for future date/time and saves as scheduled. Step 5b: Employee clicks 'Save Draft' — article saved, not published, no Karma Points awarded.                                                                                                                                                                                                                                          |
| <b>Post-condition</b>   | Article is live on the platform, visible in author's profile, community feed (if tagged), and search results. Author's Karma Points balance updated.                                                                                                                                                                                                                                                                                      |
| <b>Business Rules</b>   | 1. Title is mandatory (max 150 characters). 2. Content body is mandatory (min 100 characters). 3. At least one Category must be selected. 4. Scheduled time must be at least 30 minutes in the future.                                                                                                                                                                                                                                    |

### 6.2 UC-02: Community Admin Approves Private Community Join Request

|                         |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-02                                                                                                                                                                                                                                                                                                                                                           |
| <b>Use Case Name</b>    | Community Admin Approves/Rejects Private Community Join Request                                                                                                                                                                                                                                                                                                 |
| <b>Actor(s)</b>         | Community Admin (Primary), Requesting Employee (Passive)                                                                                                                                                                                                                                                                                                        |
| <b>Pre-condition(s)</b> | 1. Community is of type 'Private'. 2. An employee has submitted a join request. 3. Community Admin is logged in.                                                                                                                                                                                                                                                |
| <b>Basic Flow</b>       | 1. Employee submits join request for a private community. 2. Community Admin receives bell notification. 3. Admin navigates to Community Management > Join Requests. 4. Admin reviews requesting employee's profile. 5. Admin clicks 'Approve' or 'Reject'. 6. System updates membership accordingly. 7. Requesting employee receives notification of decision. |
| <b>Alternate Flow</b>   | If Admin does not act within 7 days, system sends reminder notification to Admin.                                                                                                                                                                                                                                                                               |
| <b>Post-condition</b>   | Approved: Employee gains 'Member' access to community. Rejected: Employee retains no access; reason (optional) visible to employee.                                                                                                                                                                                                                             |
| <b>Business Rules</b>   | 1. Only Community Admin or Sys Admin can approve/reject. 2. Employee may re-apply after 30 days post-rejection. 3. Reason for rejection is optional but recommended.                                                                                                                                                                                            |

### 6.3 UC-03: Employee Earns and Views Karma Points

|                         |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>      | UC-03                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Use Case Name</b>    | Employee Earns Karma Points Through Platform Activity                                                                                                                                                                                                                                                                                                                                                   |
| <b>Actor(s)</b>         | Employee (Primary), Karma Engine (System)                                                                                                                                                                                                                                                                                                                                                               |
| <b>Pre-condition(s)</b> | Employee is authenticated and performing a qualifying activity.                                                                                                                                                                                                                                                                                                                                         |
| <b>Basic Flow</b>       | 1. Employee completes a qualifying action (e.g., publishes article). 2. Karma Engine detects event. 3. Engine checks daily cap for that activity type. 4. If cap not reached: points are credited to employee's Karma balance. 5. Employee receives in-app notification of points earned. 6. If a badge threshold is crossed: badge notification is triggered. 7. Leaderboard position is recalculated. |
| <b>Alternate Flow</b>   | If daily cap is already reached: points not credited; silent pass (no error shown to user).<br>If activity is later deleted: points are not reversed.                                                                                                                                                                                                                                                   |
| <b>Post-condition</b>   | Employee's Karma balance updated. Badge awarded if threshold met. Leaderboard updated.                                                                                                                                                                                                                                                                                                                  |
| <b>Business Rules</b>   | 1. Points are non-transferable. 2. Points cannot be manually adjusted by anyone except Sys Admin via audit trail. 3. Deleted content does not result in Karma Points reversal.                                                                                                                                                                                                                          |

## 7. Integration Requirements

| Int. ID | System                    | Integration Type     | Purpose                                                                     |
|---------|---------------------------|----------------------|-----------------------------------------------------------------------------|
| INT-01  | HRMS                      | API / Database Sync  | Employee master data: Name, Designation, Dept, Location, Email, Employee ID |
| INT-02  | Active Directory / SSO    | SAML 2.0 / OAuth 2.0 | Single Sign-On authentication; role-based access provisioning               |
| INT-03  | Email Server (SMTP)       | SMTP                 | Email digest notifications, moderation alerts, system alerts                |
| INT-04  | Microsoft Stream          | Embed / Link         | Embed and stream video content hosted on Microsoft Stream                   |
| INT-05  | OneDrive                  | OneDrive Graph API   | Link to OneDrive-hosted video or document content                           |
| INT-06  | Push Notification Service | WebSocket / Push API | Real-time bell notifications within the Knome platform                      |

## 8. Analytics and Reporting

The Analytics Dashboard provides HR Administrators and System Administrators with visibility into platform engagement, content performance, and community health.

| Report ID | Report Name         | Key Metrics                                                                                               |
|-----------|---------------------|-----------------------------------------------------------------------------------------------------------|
| RPT-01    | User Engagement     | Daily Active Users (DAU), Monthly Active Users (MAU), Karma Distribution, Profile Views                   |
| RPT-02    | Community Health    | Community Members (by community), New Joins (weekly trend), Post frequency, Engagement Rate per community |
| RPT-03    | Content Performance | Posts Created, Articles Published, Video Views, Podcast Plays, Average Reactions per Content Type         |
| RPT-04    | Trending Content    | Top 10 Contributors (by Karma), Top 10 Communities (by activity), Most Viewed Posts, Most Liked Articles  |
| RPT-05    | Moderation Audit    | Reports Filed, Reports Resolved, Content Removed, User Suspensions, Moderator Actions Log                 |

## 9. Non-Functional Requirements

| NFR ID              | Category               | Requirement                                                     | Acceptance Criteria                                                       |
|---------------------|------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>NFR-PERF-01</b>  | <b>Performance</b>     | Dashboard page load time                                        | < 3 seconds at 95th percentile under 500 concurrent users                 |
| <b>NFR-PERF-02</b>  | <b>Performance</b>     | Search response time                                            | < 2 seconds for queries up to 50 characters                               |
| <b>NFR-PERF-03</b>  | <b>Performance</b>     | Video streaming initiation                                      | < 5 seconds buffering on standard 10 Mbps network                         |
| <b>NFR-Avail-01</b> | <b>Availability</b>    | Platform uptime                                                 | 99.5% monthly availability; planned downtime during off-hours             |
| <b>NFR-Avail-02</b> | <b>Availability</b>    | Disaster Recovery RTO                                           | Recovery Time Objective: < 4 hours; RPO: < 1 hour                         |
| <b>NFR-SEC-01</b>   | <b>Security</b>        | All data transmission encrypted                                 | TLS 1.2+ for all API and browser communication                            |
| <b>NFR-SEC-02</b>   | <b>Security</b>        | Authentication via SSO only; no local password                  | 100% of logins via AD/SSO; no local auth fallback                         |
| <b>NFR-SEC-03</b>   | <b>Security</b>        | Role-based access control enforced at API level                 | All endpoints return 403 for unauthorized role access                     |
| <b>NFR-SEC-04</b>   | <b>Security</b>        | Complete audit trail for content moderation and user management | All actions logged with timestamp, user ID, IP; retained for 2 years      |
| <b>NFR-SCALE-01</b> | <b>Scalability</b>     | Support concurrent active users                                 | System must handle 1000 concurrent users without degradation              |
| <b>NFR-SCALE-02</b> | <b>Scalability</b>     | Content storage scalability                                     | Support 1TB+ media storage with horizontal scalability                    |
| <b>NFR-UI-01</b>    | <b>Usability</b>       | Responsive design                                               | All pages functional on desktop (1024px+), tablet (768px), mobile (375px) |
| <b>NFR-UI-02</b>    | <b>Usability</b>       | Accessibility                                                   | WCAG 2.1 Level AA compliance for core workflows                           |
| <b>NFR-MAINT-01</b> | <b>Maintainability</b> | Modular architecture                                            | Each module independently deployable; zero downtime deployments           |

## 10. Requirements Traceability Matrix (RTM)

The following table maps each key functional requirement to its source module, associated Use Case, and test scenario for traceability through development and QA.

| Req ID   | Requirement                          | Module       | Use Case | Test Scenario                                                             |
|----------|--------------------------------------|--------------|----------|---------------------------------------------------------------------------|
| FR-DB-01 | Personalized feed display            | Dashboard    | UC-01    | Verify feed shows content from joined communities and followed users      |
| FR-DB-05 | Org-wide announcements on dashboard  | Dashboard    | —        | HR Admin posts announcement; verify visible to all employees on dashboard |
| FR-UP-01 | HRMS auto-population of profile      | User Profile | —        | New employee login shows correct Name, Dept, Designation from HRMS        |
| FR-PC-01 | Create text post (400 char max)      | Post         | —        | Post > 400 chars blocked; Post ≤ 400 chars published; Karma awarded       |
| FR-AB-01 | Article creation and publishing      | Articles     | UC-01    | Article with all fields publishes; appears in author profile and search   |
| FR-VC-01 | Video upload with metadata           | Video        | —        | MP4 ≤ 500MB uploads successfully; appears in Video Channel with metadata  |
| FR-CM-03 | Private community join approval flow | Community    | UC-02    | Join request sent; Admin notified; Approve/Reject updates membership      |
| FR-KP-01 | Karma Points auto-award              | Karma        | UC-03    | Publish article → 10 pts credited; Karma balance reflects update          |
| FR-KP-02 | Karma Leaderboard                    | Karma        | UC-03    | Top 10 contributors displayed; updates within 15 minutes of activity      |
| FR-SD-01 | Unified search across content types  | Search       | —        | Search 'digital' returns Posts, Articles, Videos, People matching keyword |

|                     |                               |                |   |                                                                    |
|---------------------|-------------------------------|----------------|---|--------------------------------------------------------------------|
| <b>FR-SM-01</b>     | URL validation before publish | Moderation     | — | Post with known malicious URL blocked; valid URL post published    |
| <b>FR-SM-05</b>     | Moderation audit trail        | Moderation     | — | Content removal action logged with timestamp, moderator ID, reason |
| <b>NFR-SEC-01</b>   | TLS 1.2+ for all traffic      | Security       | — | SSL Labs scan scores A; HTTP requests redirect to HTTPS            |
| <b>NFR-AVAIL-01</b> | 99.5% monthly uptime          | Infrastructure | — | Uptime monitoring dashboard shows SLA compliance report monthly    |

## 11. Future Enhancements (Phase 2)

| Enh. ID | Feature                          | Business Value                                                                           |
|---------|----------------------------------|------------------------------------------------------------------------------------------|
| ENH-01  | AI Content Recommendation Engine | Personalized content surfacing using ML models trained on user interaction history       |
| ENH-02  | AI Knowledge Discovery           | Surface relevant historical knowledge at the point of need; reduce knowledge re-creation |
| ENH-03  | Gamification Rewards (non-Karma) | Tangible incentives (gift cards, leave credits) linked to Karma milestones               |
| ENH-04  | Employee Recognition Programs    | Peer-to-peer recognition with public acknowledgement mechanism                           |
| ENH-05  | Learning Management Integration  | Link Knome content to LMS courses; Karma Points for course completions                   |
| ENH-06  | Event Management Module          | Create and manage internal webinars, hackathons, and town halls within the platform      |
| ENH-07  | Mentorship Network               | Structured mentor-mentee matching and session tracking within the platform               |
| ENH-08  | Live Streaming                   | Real-time broadcast of events, announcements, and Q&A sessions to all employees          |

## 12. Assumptions, Constraints, and Dependencies

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### 12.1 Assumptions

- All employees have active AD accounts provisioned prior to Knome onboarding.
- HRMS contains accurate and current employee master data (Name, Dept, Designation, Location).
- An SMTP mail server is available and configured for outbound email notifications.
- Microsoft Stream and OneDrive are available within the organizational M365 tenant for video integration.
- A push notification service (WebSocket-capable server) will be provisioned for real-time alerts.
- Organization has defined and communicated platform etiquette guidelines to employees prior to launch.

### 12.2 Constraints

- Platform must run on MPOnline's existing on-premises or approved government cloud infrastructure.
- Solution must comply with the Digital Personal Data Protection (DPDP) Act, 2023 for any personal data processed.
- All media content stored within the platform must reside on government-approved data centre infrastructure.
- Phase 1 excludes AI-driven features; all recommendation logic must be rule-based.
- External social media sharing (LinkedIn, Twitter) is not permitted for government compliance reasons.

### 12.3 Dependencies

- HRMS API availability and stability for employee master data synchronization (INT-01).
- Active Directory / SSO availability for authentication (INT-02).
- Microsoft 365 tenant access for Stream and OneDrive integration (INT-04, INT-05).
- Network infrastructure team to enable WebSocket ports for real-time notification service.

## 13. Glossary

| Term / Acronym       | Definition                                                                               |
|----------------------|------------------------------------------------------------------------------------------|
| <b>FRD</b>           | Functional Requirements Document — baseline specification for system functionality       |
| <b>Knome</b>         | Knowledge + Nome (Organization): MPOnline's enterprise social knowledge platform         |
| <b>EEP</b>           | Enterprise Excellence Portal — parent portal housing all inhouse modules including Knome |
| <b>HRMS</b>          | Human Resource Management System — source of employee master data                        |
| <b>SSO</b>           | Single Sign-On — unified authentication via Active Directory                             |
| <b>RBAC</b>          | Role-Based Access Control — access governed by user role assignment                      |
| <b>MoSCoW</b>        | Must Have / Should Have / Could Have / Won't Have — prioritization framework             |
| <b>RTM</b>           | Requirements Traceability Matrix — mapping requirements to use cases and tests           |
| <b>NFR</b>           | Non-Functional Requirement — quality attribute (performance, security, availability)     |
| <b>Karma Points</b>  | Engagement scoring system rewarding platform contributions and knowledge sharing         |
| <b>Hot Posts</b>     | Trending content ranked by engagement score algorithm                                    |
| <b>Community</b>     | A grouped discussion and collaboration space around a shared topic or department         |
| <b>Podcast</b>       | Audio-format content uploaded or recorded and published on the Knome platform            |
| <b>DPDP Act 2023</b> | Digital Personal Data Protection Act 2023 — Indian data privacy legislation              |
| <b>DAU / MAU</b>     | Daily Active Users / Monthly Active Users — platform engagement metrics                  |
| <b>TLS</b>           | Transport Layer Security — encryption protocol for data in transit                       |

— End of Document —

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